

Sonakshi Godavarthy

AI/ML Engineer | Applied AI Research | Intelligent Systems

 godavarthysonakshi@gmail.com

 +49 15158800396

 Hamburg, Germany (Open to relocate)

 portfolio-sonakshi-ten.vercel.app

 LinkedIn

 GitHub

PROFILE

AI/ML researcher and engineer with experience developing intelligent systems across knowledge graphs, multimodal document understanding, generative AI, machine learning, and data engineering. Experienced in taking research ideas from experimentation to working prototypes, including systems for aircraft maintenance, synthetic data generation, federated learning, and real-time data processing. Strong background in Python, PyTorch, cloud technologies, and scalable AI pipelines, with an interest in building reliable and interpretable AI systems for complex real-world applications.

EDUCATION

M.Sc. Data Science
Hamburg University of Technology (TUHH)
Current grade: 1.9/5.0, German scale (1.0 = highest)

Oct 2023 – Dec 2026 | Hamburg, Germany

B.Tech. Computer Science & Engineering
CMR College of Engineering & Technology
GPA: 8.87/10

Jun 2018 – May 2022 | Hyderabad, India

TECHNICAL SKILLS

AI/ML & Computer Vision
Python, PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, OpenCV, Active Learning, Federated Learning

Generative AI & Knowledge Systems
LLMs, RAG, KG-RAG, Knowledge Graphs, RDF/OWL, Ontologies, Embeddings, Semantic Search, Prompt Engineering

Data & Engineering
SQL, Pandas, NumPy, Kafka, Flink, ETL, Informatica Cloud

Cloud & Development
AWS, Azure, Docker, FastAPI, Django, React, Streamlit, REST APIs, Linux, Git

Databases & Visualization
PostgreSQL, MySQL, Snowflake, Neo4j, Power BI, Tableau, Matplotlib

PROFESSIONAL EXPERIENCE

Deutsches Zentrum für Luft- und Raumfahrt (DLR)
Master's Thesis Student

Apr 2026 – Present | Hamburg, Germany

- Developing a multimodal knowledge extraction pipeline for Airbus A320 maintenance documentation, structuring information from tables, equations, flowcharts, and technical diagrams into an ontology and knowledge graph.
- Building knowledge-grounded retrieval and reasoning methods to support accurate access to aircraft components, procedures, and maintenance relationships.
- Investigating scalable storage and retrieval architectures to support maintenance knowledge access and process analysis.

Lufthansa Technical Training GmbH
Intern - eLearning and Artificial Intelligence

Nov 2025 – Mar 2026
Hamburg, Germany

- Researched and built functional prototypes of a RAG-based document-processing pipeline supporting content summarisation, question generation, information retrieval and learner assistance.
- Designed solution architectures using DistilBART-CNN for summarisation, all-MiniLM-L6-v2 for semantic retrieval and GPT-4 for question generation and learner support.
- Studied and defined evaluation frameworks using ROUGE and Precision@K metrics to measure accuracy and retrieval relevance.

Fraunhofer Institute for Mechatronic Systems Design
Research Assistant (GenAI Incubator)

Jan 2025 – Aug 2025
Paderborn, Germany

- Developed and compared synthetic tabular-data generation methods using GPT-4 Turbo, Synthetic Data Vault (SDV), custom PyTorch diffusion models, denoising MLPs and rule-based techniques, generating 15,000+ synthetic records across experiments.
- Integrated the methods into an automated pipeline that analysed dataset columns, generated structured synthetic data and created component descriptions.
- Validated synthetic-data quality using schema checks, statistical distribution comparisons and outlier detection, with 98% of generated records passing the defined validation checks.

- Designed the interface in Figma and implemented it as a React web application, allowing users to upload datasets, select a generation model and generate additional synthetic data interactively.

Institute of Data Science Foundations (TUHH)

Feb 2025 – Jul 2025 | Hamburg, Germany

Student Assistant

- Analysed EuXFEL laboratory data to investigate why particle-beam simulations from physics-based simulation software did not always match the actual experimental measurements.
- Built a Python workflow to compare 189 real measurements with 189 API-generated simulation results using error analysis, heatmaps and t-SNE; 88.9% of the cases met the defined error threshold between simulated and measured beam-quality values.
- Investigated the 21 high-error cases and found that 86% occurred on the same measurement date, indicating possible experiment-specific or data-quality issues.

Informatica

Jul 2022 – Sep 2023 | Bangalore, India

Associate Software Engineer - Global Customer Support

- Resolved complex Informatica Cloud integration and connector issues, achieving a 95% customer satisfaction rate.
- Optimised ETL workflows (Informatica Cloud) with various database connectors, improving data pipeline efficiency and reliability.
- Developed successful technical workarounds for approximately 90% of assigned support cases and authored reusable knowledge articles to reduce repeated investigation of recurring cases.

Informatica

Aug 2021 – Jun 2022 | Bangalore, India

Intern - Global Customer Support

- Completed technical training in Informatica Intelligent Cloud Services (IICS) and worked with flat-file, Oracle, MySQL, PostgreSQL and Snowflake connectors for data integration and transformation.
- Collaborated with senior engineers to prioritise and resolve customer issues, enhancing issue response times by 20%.

PROJECTS

Applied Machine Learning & Human-Centered AI

Apr 2025 – Sep 2025

- Developed a Django-based AutoML application for CSV upload, data visualisation, model training and hyperparameter tuning.
- Built a TF-IDF and Logistic Regression sentiment classifier for 50,000 IMDB reviews, achieving 89.8% test accuracy, with active-learning strategies.
- Created an explainable AI interface using decision trees, sparse Logistic Regression and counterfactual explanations.
- Developed a movie recommender using Truncated SVD on MovieLens 32M, generating personalised Top-5 recommendations.
- Implemented a PyTorch RLHF pipeline using a CNN policy, REINFORCE, human preferences and a Bradley-Terry reward model.

Exploiting Tree-Based Topologies and Aggregation Functions Against

Oct 2024 – Jul 2025

Concept Drift in Hierarchical Federated Networks

- Investigated how Hierarchical Federated Learning systems maintain accuracy when client data changes over time (concept drift) and some clients become unavailable.
- Developed a PyTorch-based framework using a CNN on non-IID MNIST data and evaluated FedAvg, FedProx, FedNova and hybrid aggregation under synchronous, gradual and abrupt concept drift with 40% client dropout.
- Found that FedProx and FedNova adapted more effectively than FedAvg, while hybrid aggregation achieved the strongest recovery in client-dropout scenarios.

YouTube Transcript Summarisation

Aug 2024 – Aug 2024

- Developed an automated video summarisation pipeline using fine-tuned DistilBART-CNN models to generate concise summaries from YouTube transcripts.
- Integrated transcript extraction and NLP preprocessing into an automated summarisation pipeline, reducing content review time by 60%.

Real-time Traffic Monitoring

Apr 2024 – Aug 2024

- Developed a scalable streaming data pipeline to process sensor data in real-time using Kafka and Flink, enabling structured fleet monitoring dashboards.
- Applied custom operators to calculate taxi speed, distance, and performance metrics.
- Created a web interface using Streamlit to display real-time fleet information and alerts.

LANGUAGES

English — Fluent | **German** — A2, currently learning | **Hindi** — Fluent | **Telugu** — Fluent

SCHOLARSHIP

Merit Scholarship for International Students

Aug 2025

Hamburg University of Technology (TUHH)

- Awarded for academic excellence